

Pharmanovia

MY	Pharmanovia Code	739-3440
Date	21-09 -22	Version 19
Description	Bonviva 3mg 3ml MY	
Pharmanovia Previous Code		N/A

Bonviva® i.v.

Ibandronic acid

Biphosphonate – Drugs for treatment of bone diseases (M05)

1. DESCRIPTION

1.1 Therapeutic / Pharmacologic Class of Drug

Bonviva is a nitrogen-containing biphosphonate.

1.2 Type of Dosage Form

Pre-filled syringe.

1.3 Route of Administration

Intravenous

1.4 Sterile / Radioactive Statement

Sterile product.

1.5 Qualitative and Quantitative Composition

Active ingredient: ibandronic acid, monosodium salt, monohydrate Solution for injection: Each pre-filled syringe contains 3.375 mg ibandronic acid, monosodium salt, monohydrate equivalent to 3 mg of ibandronic acid in 3 ml of solution.

1.6 Product Description

Clear, colorless, solution

2. CLINICAL PARTICULARS

2.1 Therapeutic Indications

Bonviva is indicated for the treatment of postmenopausal osteoporosis, to reduce the risk of fractures. Efficacy on femoral neck fractures has not been established.
Treatment of Osteoporosis: Osteoporosis may be confirmed by the finding of low bone mass (T-score < -2.0 SD) and the presence or history of osteoporotic fracture, or a low bone mass (T-score < -2.5 SD) in the absence of documented pre-existing osteoporotic fracture.

2.2 Dosage and Administration

The recommended dose of Bonviva for treatment is 3 mg intravenous injection (administered as an intravenous injection over 15-30 seconds) every three months.

For intravenous use. Strict adherence to the intravenous route is required. Patients must receive supplemental calcium and vitamin D.
If a dose is missed, the injection should be administered as soon as convenient. Thereafter, injections should be scheduled every 3 months from the date of the last injection.

2.2.1 Special Dosage Instructions

Patients with hepatic impairment

No dosage adjustment is necessary (see section 3.2.4 Pharmacokinetics in Special Populations).

Patients with renal impairment

No dosage adjustment is necessary for patients with serum creatinine $\leq 200 \mu\text{mol/l}$ (2.3 mg/dl) or where creatinine clearance (measured or estimated) $\geq 30 \text{ ml/min}$. Bonviva 3 mg every 3 months i.v. is not recommended for use in patients who have a serum creatinine $>200 \mu\text{mol/l}$ (2.3 mg/dl) or who have a creatinine clearance (measured or estimated) $< 30 \text{ ml/min}$ because no clinical data are available from studies including such patients (see section 3.2.4 Pharmacokinetics in Special Populations).

Elderly

No dosage adjustment is necessary.

Children

Safety and efficacy have not been established in patients less than 18 years old.

2.3 Contraindications

Bonviva is contraindicated in patients with known hypersensitivity to ibandronic acid or to any of the excipients. Bonviva 3 mg every 3 months i.v. is contraindicated in patients with uncorrected hypocalcemia.

2.4 Warnings and Precautions

2.4.1 General

Bonviva like other biphosphonates administered intravenously, may cause a transient decrease in serum calcium values.
Hypocalcemia and other disturbances of bone and mineral metabolism should be effectively treated before starting Bonviva injection therapy. Adequate intake of calcium and vitamin D is important in all patients. Patients must receive supplemental calcium and vitamin D.

Bonviva 3 mg every 3 months i.v. is not recommended for use in patients who have a serum creatinine $>200 \mu\text{mol/l}$ (2.3 mg/dl) or who have a creatinine clearance (measured or estimated) $< 30 \text{ ml/min}$ because no clinical data are available from studies including such patients (see section 3.2.4 Pharmacokinetics in Special Populations).

Patients with concomitant diseases or medications which have potential for adverse effects on the kidney should be reviewed regularly in line with good medical practice during treatment.
Care must be taken not to administer Bonviva via intra-arterial administration or paravenous administration as this could lead to tissue damage. Osteonecrosis of the jaw has been reported in patients treated with biphosphonates. Most cases have been in cancer patients undergoing dental procedures, but some have occurred in patients with postmenopausal osteoporosis or other diagnoses.

Known risk factors for osteonecrosis of the jaw include a diagnosis of cancer, concomitant therapies (eg, chemotherapy, radiotherapy, corticosteroids), and co-morbid disorders (eg, anemia, coagulopathy, infection, pre-existing dental disease). Most reported cases have been in patients treated with biphosphonates intravenously but some have been in patients treated orally.
For patients who develop osteonecrosis of the jaw (ONJ) while on biphosphonate therapy, dental surgery may exacerbate the condition.

For patients requiring dental procedures, there are no data available to suggest whether discontinuation of biphosphonate treatment reduces the risk of ONJ. Clinical judgment of the treating physician should guide the management plan of each patient based on individual benefit/risk assessment.

2.4.2 Ability to Drive and Use Machines

No studies on the effects on the ability to drive and use machines have been performed.

2.4.3 Interactions with other Medicinal Products and other Forms of Interaction

Drug-Drug Interactions

In relation to disposition, no drug interactions of clinical significance are considered likely, since ibandronic acid does not inhibit the major human hepatic P450 isoenzymes and has been shown not to induce the hepatic cytochrome P450 system in rats. Furthermore, plasma protein binding is low at therapeutic concentrations and ibandronic acid is therefore unlikely to displace other drugs. Ibandronic acid is eliminated by renal excretion only and does not undergo any biotransformation. The secretory pathway appears not to include known acid or basic transport systems involved in the excretion of other drugs.

Pharmacokinetic interaction studies in postmenopausal women have demonstrated the absence of any interaction potential with tamoxifen or hormone replacement therapy (ostrogen). No interaction was observed when intravenous ibandronic acid was co-administered with melphalan/prednisolone in patients with multiple myeloma.

2.5 Use in Special Populations

2.5.1 Pregnancy

Bonviva should not be used during pregnancy and lactation.

Pregnancy

Specific studies for the 3-monthly dosing regimen have not been performed. In studies with daily i.v. dosing regimen, there was no evidence for a direct fetal toxic or teratogenic effect of ibandronic acid in rats and rabbits.

Body weight gain was decreased in F1 offspring in rats. Other adverse effects of ibandronic acid in reproductive toxicity studies in the rat were those observed with biphosphonates as a class. They include a decreased number of implantation sites, interference with natural delivery (dystocia), and an increase in visceral variations (renal pelvis ureter syndrome).
There is no clinical experience with Bonviva in pregnant women.

2.5.2 Nursing Mothers

Lactation

In lactating rats treated with 0.08 mg/kg/day i.v. ibandronic acid, the highest concentration of ibandronic acid in breast milk was 8.1 ng/ml and was seen in the first 2 hours after i.v. administration. After 24 hours, the concentration in milk and plasma was similar, and corresponded to about 5% of the concentration measured after 2 hours.
It is not known whether Bonviva is excreted in human milk.

2.6 Undesirable Effects

2.6.1 Clinical Trials

Treatment of postmenopausal osteoporosis

2.5 mg daily oral dosing

The safety of Bonviva 2.5 mg daily was evaluated in 1251 patients treated in 4 placebo-controlled clinical studies; 73% of these patients came from the pivotal three-year treatment study (MF 4411). The overall safety profile of Bonviva 2.5 mg daily in all these studies was similar to that of placebo. The overall proportion of patients who experienced an adverse drug reaction, i.e. adverse event with a possible or probable relationship to trial medication, in the pivotal treatment study (MF 4411) was 19.8% for Bonviva and 17.9% for placebo.

3 mg every 3 months i.v. dosing

In the pivotal two-year study in postmenopausal women with osteoporosis (BM16550) the overall safety of intravenous injection of Bonviva 3 mg every 3 months and oral Bonviva 2.5 mg daily were shown to be similar. The overall proportion of patients who experienced an adverse drug reaction was 26.0 % and 28.6 % for Bonviva 3 mg injection every 3 months and 20.4 % and 22.6 % for oral Bonviva 2.5 mg daily after one year and two years, respectively. The majority of adverse drug reactions were mild to moderate in intensity. Most cases of adverse drug reactions did not lead to cessation of therapy.
Table 1 and table 2 lists adverse events after one year and two years of treatment, respectively from the pivotal phase III trial BM16550, reported as possibly or probably related to trial medicinal product, in more than 1% of the patients treated with either intravenous injection of Bonviva 3 mg every 3 months or intravenous injection of placebo plus oral Bonviva 2.5 mg daily. Adverse drug reactions reported in patients treated with Bonviva 3 mg injection at a frequency equal to or less than that in orally treated patients, are not included.

Manufacturer	WAYMADE		
CMO Code	739-3440		
Previous Ref.	6031644-L		
Size	580x360mm		
Tech. Drawing	BONVIVA 3MG 3ML 580X360 CORRECT FOLDING LEAFLET GUIDE		
Laetus Code	N/A		
Edge Marks	N/A		
Min. Font Size	8 pt	Min. Code Size	7 pt
Notes			

Tables 1 and 2 also show adverse drug reactions in patients treated for 3 years with oral Bonviva 2.5 mg daily in the anti-fracture study (MF4411). For both studies, adverse drug reactions are listed which occurred with a higher incidence in Bonviva-treated patients compared with the placebo-treated patients of study MF4411. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Table 1: Common adverse drug reactions ($>1/100$, $\leq 1/10$) in the phase III osteoporosis study BM16550 after one year of treatment and in the phase III anti-fracture study MF 4411 (three-year study), that were considered by the investigator to be possibly or probably related to study medicinal product.

	One year data in study BM 16550		Three year data in study MF 4411	
System Organ Class / Adverse drug reaction	Bonviva injection 3 mg every 3 months (N=469) ADR No. (%)	Placebo injection 1 oral Bonviva 2.5 mg daily (N=465) ADR No. (%)	Oral Bonviva 2.5 mg daily (N=977) ADR No. (%)	Placebo (N=975) ADR No. (%)
Gastrointestinal disorders				
Gastritis	5 (1.1)	4 (0.9)	7 (0.7)	5 (0.5)
Diarrhoea	5 (1.1)	2 (0.4)	14 (1.4)	10 (1.0)
Abdominal pain	13 (2.8)	15 (3.2)	21 (2.1)	28 (2.9)
Dyspepsia	12 (2.6)	18 (3.9)	40 (4.1)	26 (2.7)
Nausea	8 (1.7)	12 (2.6)	18 (1.8)	22 (2.3)
Constipation	5 (1.1)	7 (1.5)	3 (0.3)	9 (0.9)
Musculoskeletal disorders				
Musculo-skeletal pain	5 (1.1)	2 (0.4)	-	-
Arthralgia	11 (2.3)	4 (0.9)	4 (0.4)	4 (0.4)
Myalgia	8 (1.7)	4 (0.9)	18 (1.8)	8 (0.8)
General System disorders				
Influenza like illness*	22 (4.7)	4 (0.9)	3 (0.3)	2 (0.2)
Fatigue	5 (1.1)	2 (0.4)	3 (0.3)	4 (0.4)
Nervous system disorders				
Headache	5 (1.1)	3 (0.6)	8 (0.8)	6 (0.6)
Skin disorders				
Rash	4 (0.9)	3 (0.6)	12 (1.2)	7 (0.7)

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* Transient, influenza-like symptoms have been reported in patients receiving intravenous injection of Bonviva 3 mg every 3 months, typically in association with the first dose. Influenza-like illness includes events reported as acute phase reaction or symptoms, including myalgia, arthralgia, fever, chills, fatigue, nausea, loss of appetite, and bone pain. Such symptoms were generally of short duration, mild or moderate in intensity, and resolved during continuing treatment without requiring remedial measures.

Table 2: Common adverse drug reactions ($>1/100$, $\leq 1/10$) in the phase III osteoporosis study BM16550 after two years of treatment (cumulative data) and in the phase III anti-fracture study MF 4411 (three-year study), that were considered by the investigator to be possibly or probably related to study medicinal product.

	Two year data in study BM 16550		Three year data in study MF 4411	
System Organ Class / Adverse drug reaction	Bonviva injection 3 mg every 3 months (N=469) ADR No. (%)	Placebo injection 1 oral Bonviva 2.5 mg daily (N=465) ADR No. (%)	Oral Bonviva 2.5 mg daily (N=977) ADR No. (%)	Placebo (N=975) ADR No. (%)
Gastrointestinal disorders				
Gastritis	6 (1.3)	4 (0.9)	7 (0.7)	5 (0.5)
Diarrhoea	5 (1.1)	3 (0.6)	14 (1.4)	10 (1.0)
Abdominal pain	17 (3.6)	21 (4.5)	21 (2.1)	28 (2.9)
Dyspepsia	14 (3.0)	19 (4.1)	40 (4.1)	26 (2.7)
Nausea	8 (1.7)	13 (2.8)	18 (1.8)	22 (2.3)
Constipation	5 (1.1)	7 (1.5)	3 (0.3)	9 (0.9)
Musculoskeletal disorders				
Musculo-skeletal pain	5 (1.1)	2 (0.4)	-	-
Arthralgia	13 (2.8)	4 (0.9)	4 (0.4)	4 (0.4)
Myalgia	8 (1.7)	4 (0.9)	18 (1.8)	8 (0.8)
Back Pain	5 (1.1)	1 (0.2)	3 (0.3)	2 (0.2)
General System disorders				
Influenza like illness*	21 (4.5)	4 (0.9)	3 (0.3)	2 (0.2)
Fatigue	5 (1.1)	2 (0.4)	3 (0.3)	4 (0.4)
Nervous system disorders				
Headache	6 (1.3)	3 (0.6)	8 (0.8)	6 (0.6)
Skin disorders				
Rash	4 (0.9)	4 (0.9)	12 (1.2)	7 (0.7)

MedDRA version 8.0

* Transient, influenza-like symptoms have been reported in patients receiving intravenous injection of Bonviva 3 mg every 3 months, typically in association with the first dose. Influenza-like illness includes events reported as acute phase reaction or symptoms, including myalgia, arthralgia, fever, chills, fatigue, nausea, loss of appetite, and bone pain. Such symptoms were generally of short duration, mild or moderate in intensity, and resolved during continuing treatment without requiring remedial measures.

Adverse drug reactions occurring at a frequency of $\leq 1\%$ in study BM16650 that were considered by the investigator to be possibly or probably related to study medicinal product:

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Uncommon ($\geq 1/1,000$ and $\leq 1/100$)

Musculoskeletal disorders: Bone pain

General disorders and administration site conditions: Asthenia , Injection site reactions

Vascular Disorders: Phlebitis/thrombophlebitis

Rare ($\geq 1/10,000$ and $\leq 1/1,000$)

Immune system disorders: Hypersensitivity reactions

Skin and subcutaneous tissue disorders: Angioedema , Facial swelling/ oedema , Urticaria

2.6.1.1 Laboratory Abnormalities

There was no evidence that Bonviva 3 mg every 3 months i.v.induced laboratory abnormalities indicative of hepatic or renal dysfunction, impaired hematologic system, hypocalcemia or hypophosphatemia

2.6.2 Post Marketing

Musculoskeletal and connective tissue disorders :

Osteonecrosis of the jaw has been reported very rarely in patients treated with ibandronic acid (see 2.4. Warnings and Precautions).

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2.7 Overdose

No specific information is available on the treatment of overdose with Bonviva.

Based on knowledge of this class of compounds, intravenous overdose may result in hypocalcemia, hypophosphatemia, and hypomagnesaemia. Clinically relevant reductions in serum levels of calcium, phosphorus, and magnesium should be corrected by intravenous administration of calcium gluconate, potassium or sodium phosphate, and magnesium sulfate, respectively.

3. PHARMACOLOGICAL PROPERTIES AND EFFECTS

3.1 Pharmacodynamic Properties

The pharmacodynamic action of ibandronic acid is inhibition of bone resorption. In vivo, ibandronic acid prevents experimentally induced bone destruction caused by cessation of gonadal function, retinoids, tumors or tumor extracts. In young (fast growing) rats, the endogenous bone resorption is also inhibited, leading to increased bone mass compared with untreated animals.
Animal models confirm that ibandronic acid is a highly potent inhibitor of osteoclastic activity. In growing rats, there was no evidence of impaired mineralization even at doses greater than 5,000 times the dose required for osteoporosis treatment.

The high potency and therapeutic margin of ibandronic acid allows for more flexible dosing regimens and intermittent treatment with long drug-free intervals at comparatively low doses.

Both daily and intermittent (with prolonged dose-free intervals) long-term administration in rats, dogs and monkeys were associated with formation of new bone of normal quality and/or increased mechanical strength even in doses in excess of any pharmacologically intended dose, including the toxic range. In humans, the efficacy of both daily and intermittent administration with a dose-free interval of 9-10 weeks of ibandronic acid was confirmed in a clinical trial (MF 4411), in which Bonviva demonstrated anti-fracture efficacy. Both daily and intermittent (with a drug-free interval of 9-10 weeks per quarter) oral doses as well as intravenous doses of Bonviva in postmenopausal women produced biochemical changes indicative of dose-dependent inhibition of bone resorption. Bonviva intravenous injection decreased levels of serum CTX within 3-7 days of starting treatment and decreased levels of osteocalcin within 3 months.

Following treatment discontinuation, there is a reversion to the pathological pre-treatment rates of elevated bone resorption associated with postmenopausal osteoporosis.

The histological analysis of bone biopsies after two and three years of treatment of postmenopausal women with doses of Bonviva 2.5 mg daily and intermittent i.v. doses of up to 1 mg every 3 months showed bone of normal quality and no indication of a mineralisation defect.

3.1.1 Mechanism of Action

Ibandronic acid is a highly potent biphosphonate belonging to the nitrogen-containing group of biphosphonates, which act on bone tissue and specifically inhibit osteoclast activity. It does not interfere with osteoclast recruitment. The selective action of ibandronic acid on bone tissue is based on the high affinity of this compound for hydroxyapatite, which represents the mineral matrix of the bone. Ibandronic acid reduces bone resorption, with no direct effect on bone formation. In postmenopausal women, it reduces the elevated rate of bone turnover towards premenopausal levels, leading to a progressive net gain in bone mass. Daily or intermittent administration of ibandronic acid results in reduced bone resorption as reflected in reduced levels of serum and urinary biochemical markers of bone turnover, increased BMD and a decreased incidence of fractures.

3.1.2 Clinical / Efficacy Studies

Treatment of postmenopausal osteoporosis

Bonviva 2.5 mg daily

A statistically significant and medically relevant decrease in the incidence

of new radiographic morphometric and clinical vertebral fractures was demonstrated in a 3-year, randomized, double-blind, placebo-controlled, fracture study (MF 4411). Bonviva was evaluated at oral doses of 2.5 mg daily and 20 mg intermittently (20 mg every other day for 12 doses at the start of each 3-month cycle, followed by a 9-10 week drug-free interval). Bonviva was taken 60 minutes before the first food or drink of the day (post-dose fasting period). The study enrolled 2,946 women aged 55 to 80 years (2,928 were eligible for efficacy), who were at least 5 years postmenopausal, who had a lumbar spine BMD 2.5 SD below the premenopausal mean (T-score) in at least one vertebra [L1-L4], and who had one to four prevalent vertebral fractures. All patients received 500 mg calcium and 400 IU vitamin D daily.

Bonviva showed a statistically significant and medically relevant reduction in the incidence of new vertebral fracture with both regimens tested.

The 2.5 mg daily regimen reduced the occurrence of new radiographic vertebral fractures by 62% over the three year duration of the study. Clinical vertebral fractures were also reduced by 49%. The strong effect on vertebral fractures was furthermore reflected by a statistically significant reduction of height loss compared to placebo. The anti-fracture effect was consistent over the duration of the study. There was no indication of a waning of the effect over time.

Although the clinical fracture trial for ibandronic acid was not specifically designed to demonstrate fracture efficacy in non-vertebral fractures, a relative risk reduction of similar magnitude (69%) as demonstrated for vertebral fractures was observed for non-vertebral fractures in a subgroup of patients being at higher fracture risk (femoral neck BMD T - score < -3.0 SD).

The observation of non-vertebral fracture efficacy in high-risk subgroup is consistent with clinical trial findings for other biphosphonates.

Three-year lumbar spine BMD increase compared to placebo was 5.3% for the daily regimen. Compared to baseline this increase was 6.5%. Biochemical markers of bone turnover (such as urinary CTX and serum osteocalcin) showed the expected pattern of suppression to premenopausal levels and reached maximum suppression within a period of 3-6 months. A clinically meaningful reduction of 50% and 78 % of biochemical markers of bone resorption was observed as early as one month after start of treatment with Bonviva 2.5 mg daily and 20 mg intermittently, respectively. Decreases in biochemical markers of bone resorption were evident within 7 days of starting treatment. Bonviva 3 mg injection every 3 months

Bone mineral density (BMD)

Bonviva 3 mg intravenous injection, administered every 3 months, was shown to be at least as effective as oral Bonviva 2.5 mg daily in a 2-year, randomised, double-blind, multicentre, non-inferiority study (BM16550) of postmenopausal women (1386 women aged 55 - 80) with osteoporosis (lumbar spine BMD T-score below -2.5 SD at baseline). All patients received 400 IU vitamin D and 500 mg calcium supplementation per day. This was demonstrated in both the primary analysis at one year and in the confirmatory analysis at two years endpoint (Table 3).

The primary analysis of data from study BM16550 at one year and the confirmatory analysis at 2 years demonstrated the non-inferiority of 3 mg every 3 months injection dosing regimen compared to 2.5 mg oral daily dosing regimen, in terms of mean increases in BMD at lumbar spine, total hip, femoral neck and trochanter (Table 3).

Table 3: Mean relative change from baseline of lumbar spine, total hip, femoral neck and trochanter BMD after one year (primary analysis) and two years of treatment (Per-Protocol Population) in study BM 16550.

	One year data in study BM 16550		Two year data in study BM 16550	
Mean relative changes from baseline % [95% CI]	Bonviva 2.5 mg daily (N=377)	Bonviva 3 mg injection every 3 month (N=365)	Bonviva 2.5 mg daily (N=334)	Bonviva 3 mg injection every 3 months (N=334)
Lumbar spine L2-L4 BMD	3.8(3.4, 4.2)	4.8(4.5, 5.2)	4.8(4.3, 5.4)	6.3(5.7, 6.8)
Total hip BMD	1.8(1.5, 2.1)	2.4(2.0, 2.7)	2.2(1.8, 2.6)	3.1(2.6, 3.6)
Femoral neck BMD	1.6(1.2, 2.0)	2.3(1.9, 2.7)	2.2(1.8, 2.7)	2.8(2.3, 3.3)
Trochanter BMD	3.0(2.6, 3.4)	3.8(3.2, 4.4)	3.5(3.0, 4.0)	4.9(4.1, 5.7)

Furthermore, Bonviva 3 mg injection every 3 months was proven superior to oral Bonviva 2.5 mg daily for increases in lumbar spine BMD in a prospectively planned analysis at one year, $p<0.001$, and at two years, $p<0.001$.

For lumbar spine BMD, 92.1 % of patients receiving 3 mg injection every 3 months increased or maintained their BMD after 1 year of treatment (i.e. were responders) compared with 84.9 % of patients receiving oral 2.5 mg daily ($p<0.002$). After 2 years of treatment, 92.8 % of patients receiving 3 mg injections and 84.7 % of patient receiving 2.5 mg oral therapy had increased or maintained lumbar spine BMD. For total hip BMD, 82.3 % of patients receiving 3 mg injection every 3 months were responders at one year, compared with 75.1 % of patients receiving 2.5 mg daily orally ($p<0.02$). After 2 years of treatment, 85.6 % of patients receiving 3 mg injections and 77.0 % of patient receiving 2.5 mg oral therapy had increased or maintained total hip BMD ($p<0.004$). The proportion of patients who increased or maintained their BMD at one year at both lumbar spine and total hip was 7

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Description

Bonviva 3mg 3ml MY

Pharmanovia Previous Code

N/A

Manufacturer	WAYMADE		
CMO Code	739-3440		
Previous Ref.	6031644-L		
Size	580x360mm		
Tech. Drawing	BONVIVA 3MG 3ML 580X360 CORRECT FOLDING LEAFLET GUIDE		
Laetus Code	N/A		
Edge Marks	N/A		
Min. Font Size	8 pt	Min. Code Size	7 pt
Notes			

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